

Sprite Interactions

Learning Objectives

- Create multiple sprites that interact with each other
- Use broadcasts to coordinate sprite communication
- Design scenarios where sprites work together to complete tasks

Engage (5-7 minutes)

In a village, people work together: the farmer ploughs, the wife sows, children chase birds. Each has a role, and they coordinate through words and signals. Sprite interactions in Scratch teach how multiple characters communicate and work together, like villagers coordinating during harvest season.

Explore (10-12 minutes)

Students create two sprites and explore how they can interact: following each other, changing when they touch, or responding to the same events. Like farmers in adjacent fields communicating across boundaries, sprites share information through broadcasts and direct sensing.

Explain (10-12 minutes)

Sprites interact through several methods: direct sensing (touching another sprite), broadcasts (messages sent between sprites), and shared variables (common data). Broadcasts are like village drums — one sprite sends a message, and all sprites listening respond accordingly, coordinating their actions.

Elaborate (8-10 minutes)

Students create a simple scene where a cat sprite chases a mouse sprite. When they touch, both sprites broadcast messages to change their costumes. Like a chase game in the village square, sprites interact through movement, sensing, and communication.

Evaluate (5-8 minutes)

Can students create at least two sprites that interact? Can they implement broadcast communication between sprites? Do they understand how sprites can coordinate their actions?